Throughput of Tiered Document Parsing

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Abstract. We compare four parsing tiers on a small synthetic corpus and report how layout analysis, optical character recognition and formula recognition change throughput and fidelity.

1. Introduction

Let $x_i \in \mathbb{R}^d$ denote the feature vector of page i, and let $\alpha \in (0, 1)$ be a mixing weight. The empirical loss is $\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell(f_\theta(x_i), y_i)$ and the regulariser satisfies $\|\theta\|_2 \le 1$.[1] The latency model that we fit is given by

$$T(n) = a + b\,n + c\,n^2, \qquad \int_0^{\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2} \tag{1}$$

where n is the number of pages and a, b, c are fitted constants. In Section 2 we report the measured latencies, and we discuss the Hungarian-language test page in Section 3.

2. Method

Each document is parsed once per tier. The median and the 95th percentile of the wall-clock time are recorded for every run, and the results are summarised in Table 1 on the next page.

[1] The norm constraint is the usual unit-ball regularisation; it is stated here only as an example footnote.

Table 1. Parsing latency by tier (milliseconds).

Tier	Pages	Median	p95
flash	12	810	1 240
basic	12	1 126	2 980
standard	12	4 310	9 870
advanced	12	7 450	15 200

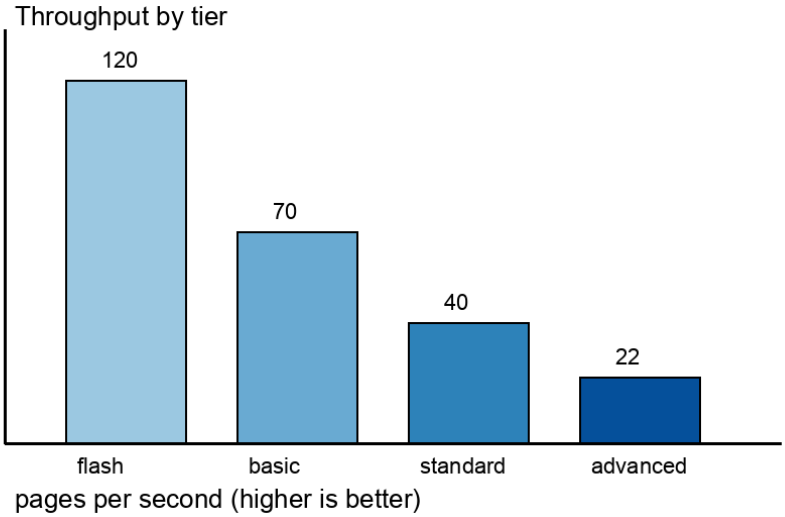


Figure 1. Relative throughput by tier.

3. Hungarian test page

Az árvíztűrő tükörfúrógép egy magyar pangram: minden ékezetes magyar betűt tartalmaz, így az ő és az ű betűt is. A dokumentum feldolgozása után a szövegnek pontosan meg kell maradnia. ÁRVÍZTŰRŐ TÜKÖRFÚRÓGÉP.

The quadratic term satisfies $c > 0$ for every tier, and the fitted curve passes within $\epsilon = 0.05$ of each measurement.